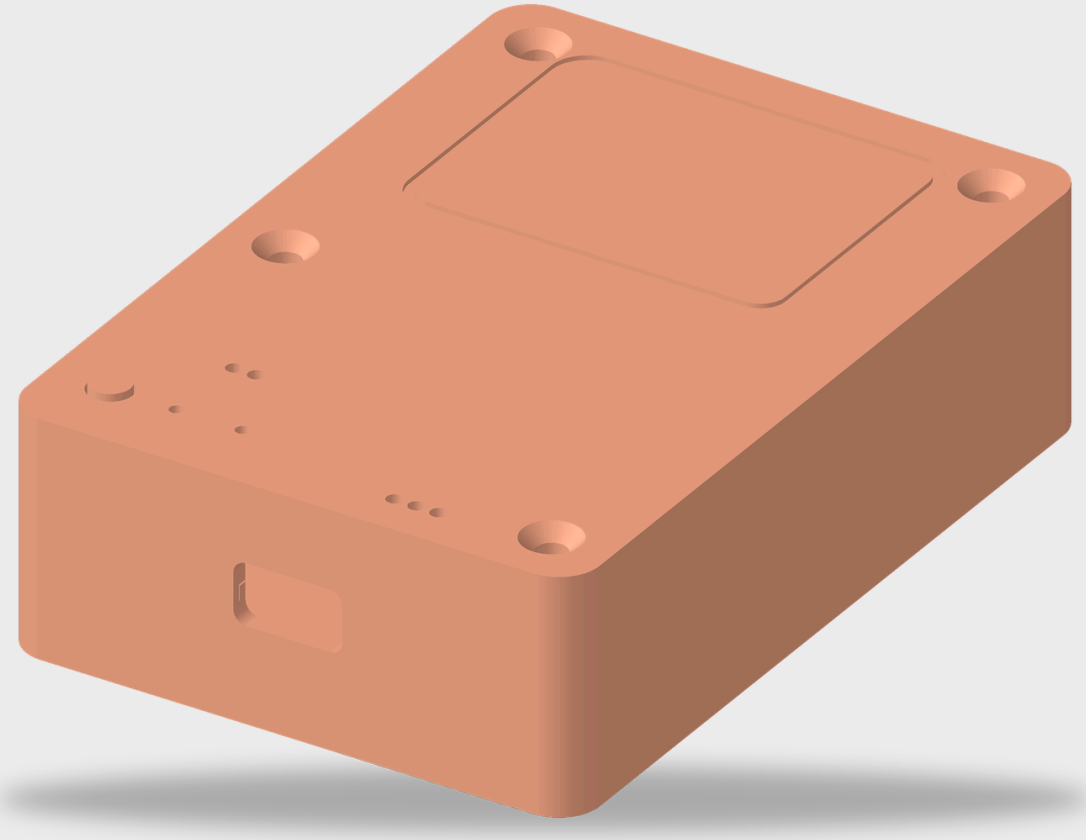
Attendance Management System

Group 05

Rajinthan R. : 220502M
Praveen V.V.J. : 220491B
Sulojan R. : 220626V

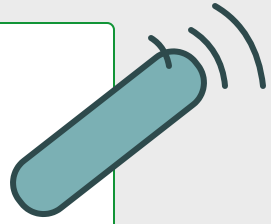


How System Works

- Automate the process of taking attendance in live physical classes.
- A handheld NFC reader records each student card and the time.

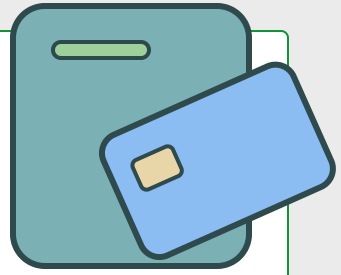
01

Pass the reader
hold it near the cards



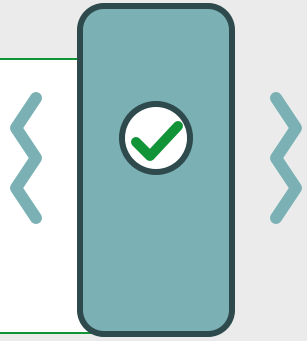
02

Tap the student ID



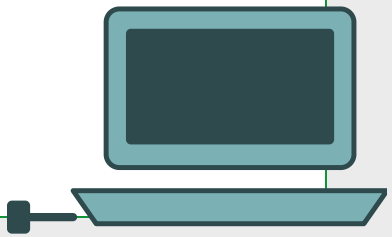
03

Instant feedback
light and buzz



04

Plug in over USB
to retrieve the records



Justification of NFC

- Student ID cards and NFC tags are usually **13.56 MHz ISO 14443-A** (MIFARE, NTAG). One reader IC, the **ST25R3916**, reads them over SPI.
- Every card has its own number (4, 7 or 10 bytes). That number is all the device stores.
- The reader can **look for a card by itself** every 100 ms while the MCU sleeps.
- The antenna is a **3-turn loop etched on the PCB**: no extra part.

	First design 125 kHz EM4100	Final 13.56 MHz NFC
Cards	EM4100 tags	ISO 14443-A cards
Reader	discrete circuit	one chip
Card detect	separate touch IC	reader's own wake-up

18.2 mA
polling the reader all the time

194 µA
with the reader's wake-up mode

Measured on the board, 8 October 2026 (polling figure from the model).

Power Analysis

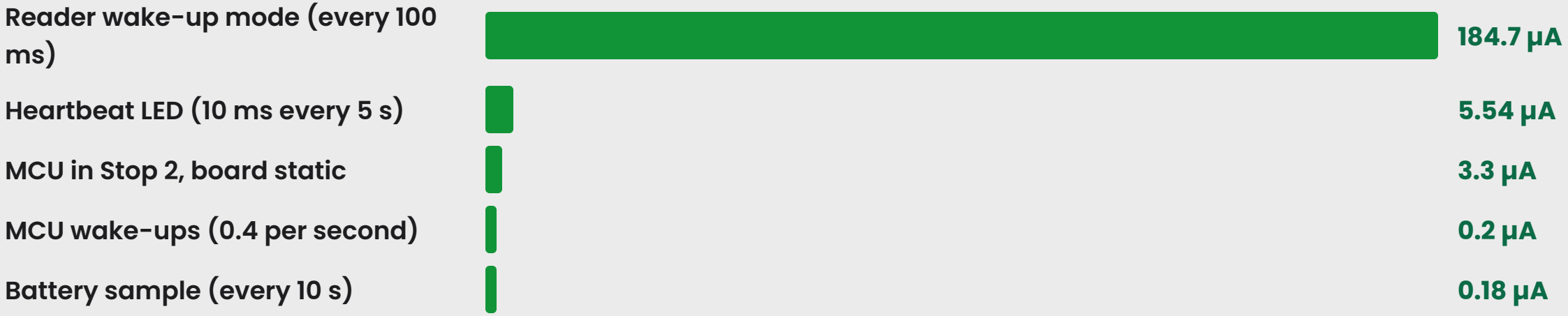
1.8 µA
switched off (Standby)

194 µA
idle, waiting for a card

2.5 µAh
one card tap (8.9 mC)

18 months
on a 1000 mAh cell (13 worst case)

Where the 194 µA goes

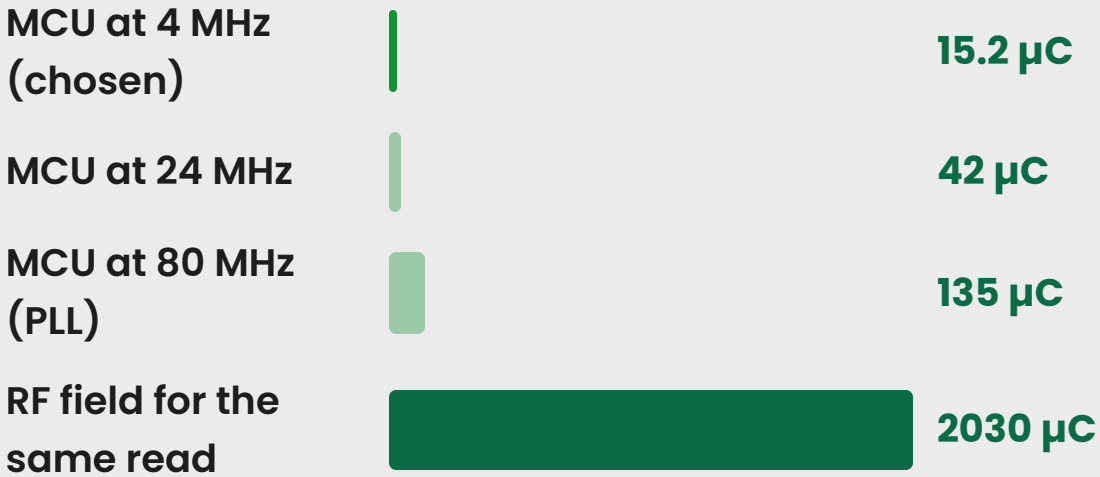


Measured with a meter in series with the cell (power-test build). Battery life assumes 4 lectures a day of 100 students, with self-discharge. The RF field (203 mA) and the motor (56.5 mA) only run for milliseconds per tap.

Clock Speed Calculation

Clock	Speed	Why
Scanning	**4 MHz (MSI)**	**no PLL to relock on each wake-up**
USB session	24 MHz (MSI)	USB needs above 14.2 MHz; power comes from the cable
USB clock	HSI48 + CRS	locks to the host, no crystal
RTC, timers	32.768 kHz	runs through sleep for under 1 µA
Reader SPI	500 kHz	chip allows 10 MHz

Charge in one 15 ms card read



Reading a card waits on the radio, not on the processor, so the slowest clock that meets the timing wins. At 4 MHz the MCU is under 1 % of a read.

Q = I × t: 1.01 mA × 15 ms at 4 MHz (measured); 2.8 mA and 9 mA from the datasheet for 24 and 80 MHz.

Memory

STM32L432KB: 128 kB of flash, 64 pages of 2 kB

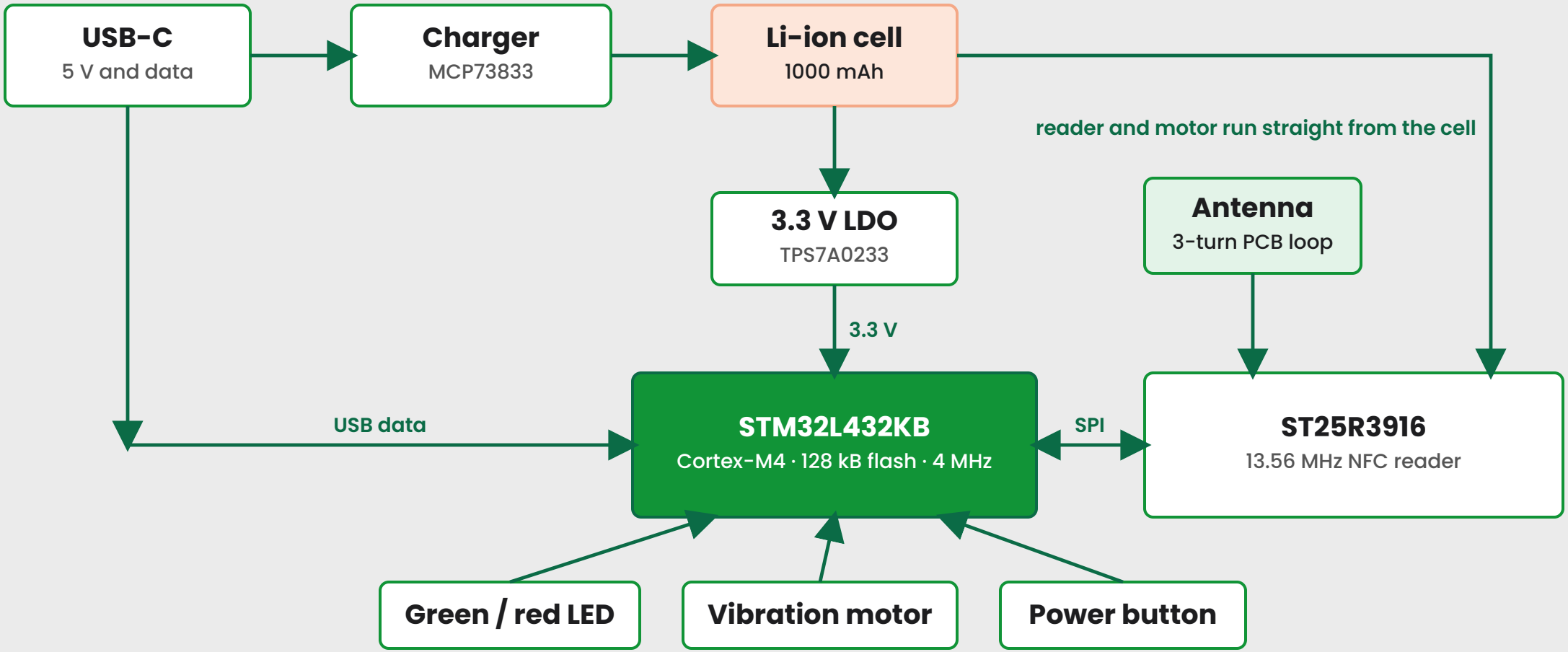


- One record is **8 bytes**: card number and time. That is exactly one flash double-word.
- A page holds 254 records. **6,858 records** is about **64 lectures** of 100 students.
- The log fills in about 16 teaching days. The app clears it each time a lecture starts.

Build	Image	Fits 72 kB
Debug (-Og)	62.1 kB	yes, 11.4 kB free
Release (-Os)	**51.5 kB**	**yes, 21.7 kB free**

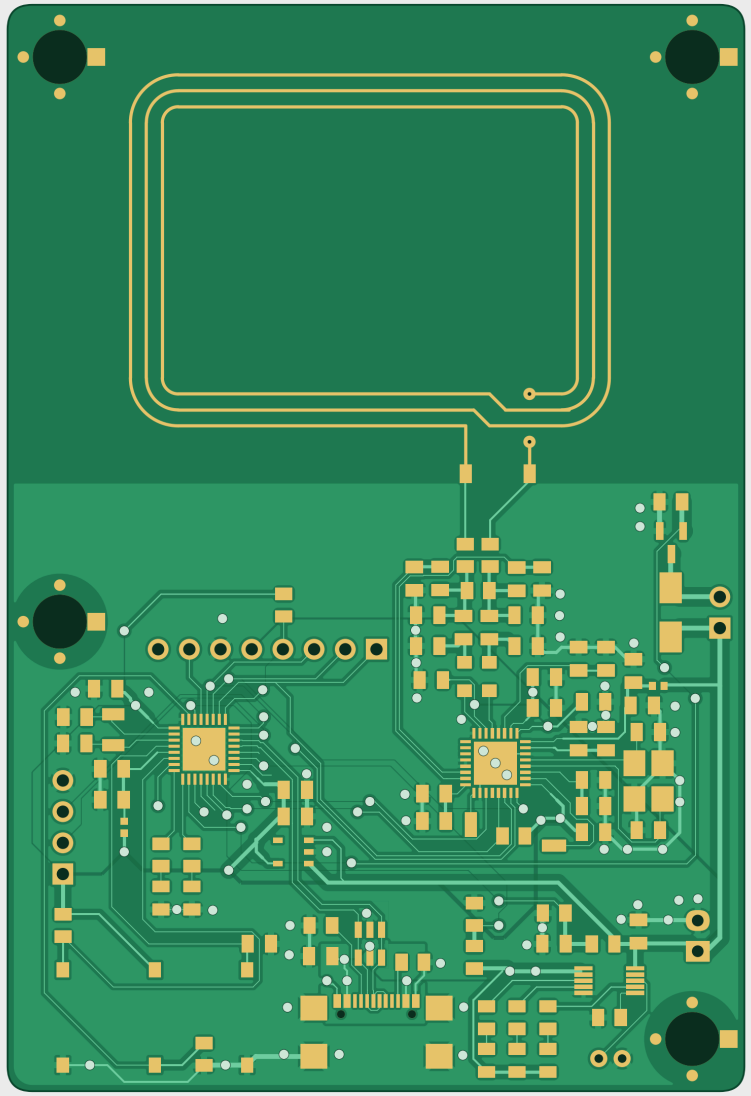
Every page given to the program is 254 records less in the log, so 36 / 28 is the split we chose.

Architecture



Firmware is two layers: **App** decides (it also runs on a PC for testing), **Bsp** only talks to the hardware. The USB drive holds ATTEND.CSV (taps), LECTURES.CSV, STATUS.TXT and SETTINGS.CSV.

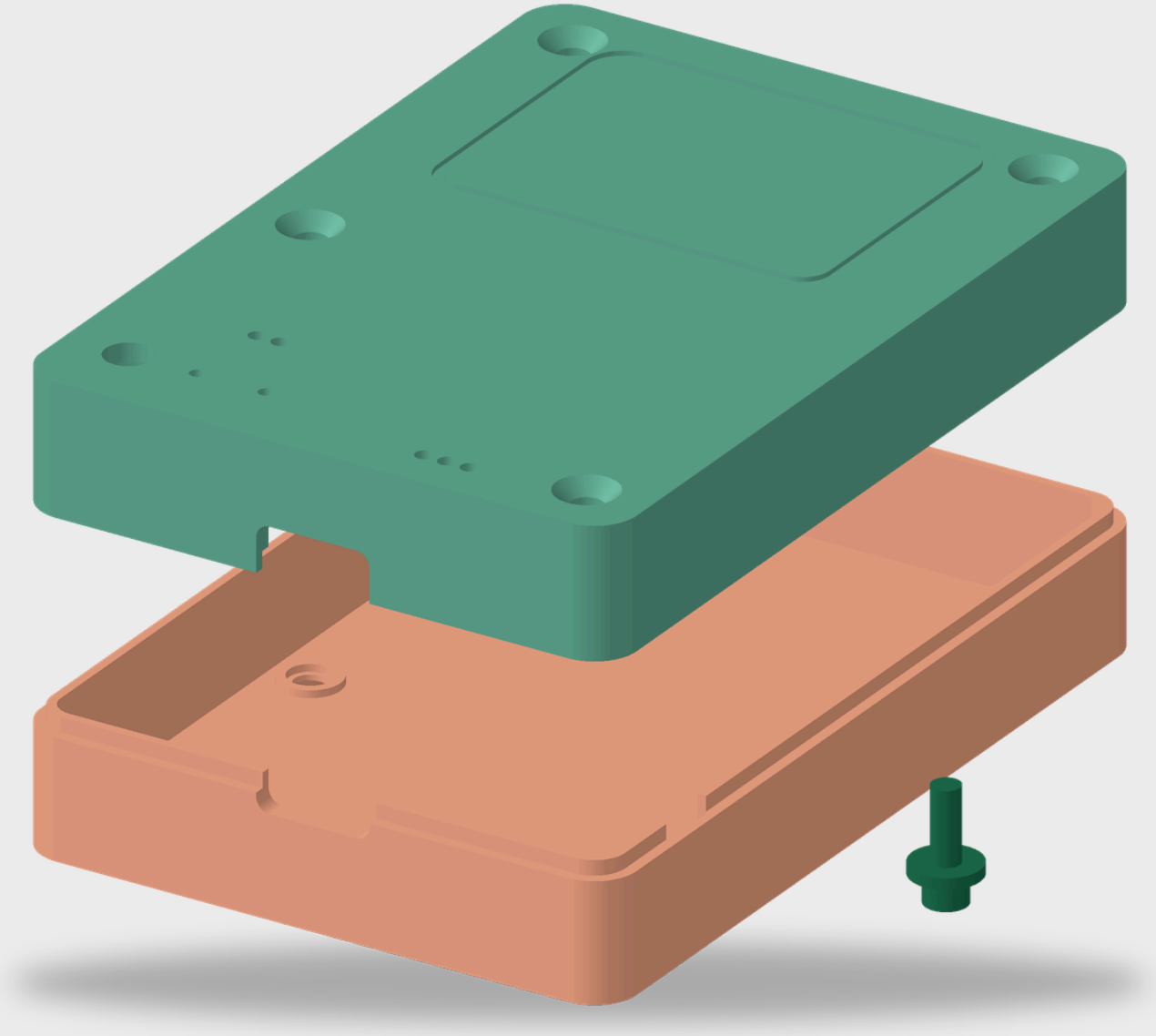
PCB Design



- Two layers, **60 × 89 mm**, 88 parts, all on the top side.
- NFC loop antenna etched on the board, **40 × 30 mm**, 3 turns.
- USB-C at the bottom edge charges the cell and shows the drive.
- Reader and motor run straight from the cell; the MCU from a 3.3 V LDO.

Block	Part
MCU	STM32L432KB
NFC reader	ST25R3916
Charger	MCP73833
LDO	TPS7A0233
USB protection	USBLC6-2SC6

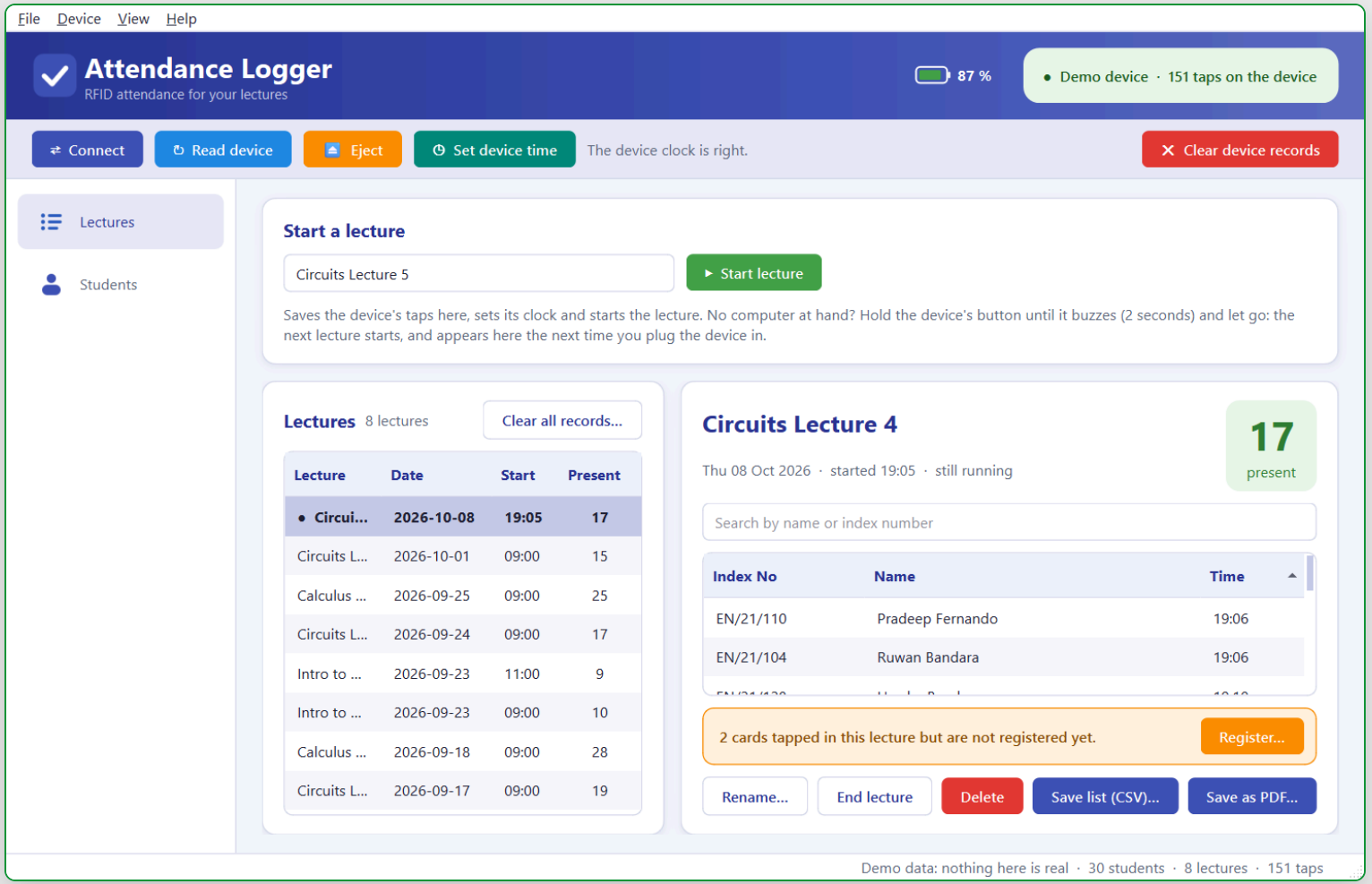
Enclosure Design



- **Two printed parts and a button plunger: base, lid, power button.**
- **Size 65 × 94 × 27.5 mm.**
- **The board screws down on four standoffs; the lid takes four M3 screws.**
- **A bay under the board holds a LiPo up to 45 × 70 × 7 mm.**
- **Cut-outs for the USB-C, the LEDs and the buttons.**

Parametric: **Enclosure/enclosure.py** reads the board outline and writes the STL files.

The Companion App



- **Plug the device in and the app reads every tap.**
- **Type a lecture name and press Start lecture.**
- **See who came: index number, name, time.**
- **Save each lecture as CSV or PDF.**
- **Register new cards from the device.**

Python + Qt, with a SQLite database on the PC. The device keeps no student list.

Quick Guide

You see	It means
🟢 Green + one short buzz	Card recorded
🟢 Two quick buzzes	Already recorded in this lecture
🟢 Green + three quick buzzes	New lecture started
🔴 Red blinking 5 times	Battery low
🔴 Red + four buzzes	Memory full, tap not recorded
🔴 Red + long buzz	Switching off

The power button

Press (off)	Switch on
Tap	Show the battery
Hold 2 s	Start next lecture
Hold 5 s	Switch off

It switches itself off after 3 minutes without a tap. Records are kept.

1 Switch on

2 Tap cards

3 Hold 2 s for the next lecture

4 Plug in, open the app

Total Cost

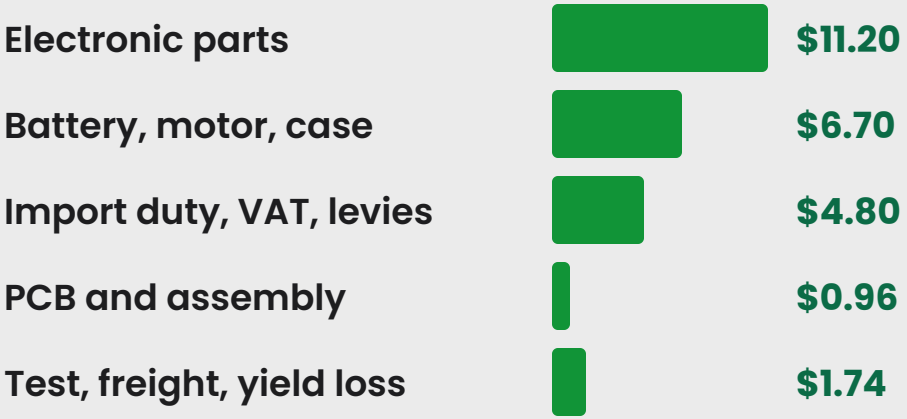
Units made	1	100	1,000	10,000
Cost per unit (USD)	227.62	32.43	**25.40**	18.31
Cost per unit (LKR)	75,115	10,702	**8,382**	6,042
Price at 40 % margin (USD)	**379**	**54.05**	**42.33**	**30.52**

Landed in Sri Lanka and tested: parts from LCSC, boards from JLCPCB, freight, 25 % import charges, final assembly. 1 USD = 330 LKR.

Biggest parts at 1,000 units

STM32L432KB $3.25 · ST25R3916 $3.40 · charger $1.12 · USB-C $0.72

Where $25.40 goes (1,000 units)



Up to 100 units the case is 3D printed. At 1,000 it is moulded with an aluminium tool ($3,000), about $3 a unit.

What Is Next

- Tune the antenna (C16 to C19, R15), then measure the field current and the idle current again.
- Count false wake-ups over an hour with no card near.
- Ask ST or a distributor for a quote at 100 units or more: LCSC held only 27 ST25R3916.
- Radio type approval for a 13.56 MHz transmitter (TRCSL) before selling.
- Change R6 to 2 kΩ so the charger suits a USB-C port (500 mA).

Thank you